Cisco ASA Firewall

Cisco Packet Tracer - Step-by-Step Build Guide (Project 5)

Goal: a client business is worried about the security of its office network, so a dedicated firewall is installed between the Office network and the Internet. The Cisco ASA separates an inside (trusted) network from an outside (untrusted) network, takes over address translation (PAT) from the Edge Router, and inspects ICMP so that replies to the PCs' pings are allowed back in.

You will use: 2 Office PCs, 1 Access Switch, 1 Edge Router, 1 ISP Router, 1 Public Server (simulated Internet), 1 ASA Firewall (its Ethernet ports work like switch ports, as on the ASA 5505).

Three parts: Part 1 reviews and tests the current network (before the firewall). Part 2 adds the ASA and changes the Edge Router and the ISP Router. Part 3 configures the ASA.

1. What you are building

Area	Devices	Network
Office network (inside)	2 Office PCs + Access Switch	192.168.1.0/24
Firewall	ASA-FW between the Edge Router and the Access Switch	inside (level 100) / outside (level 0)
Edge Router - ASA link	Edge Router G0/1 + ASA E0/0	209.165.200.224/29
Edge Router - ISP link	Edge Router G0/0 + ISP Router	203.0.113.0/30
Simulated Internet	ISP Router + Public Server	-

Note: Before you change anything, review the network diagram, the IP addressing plan, the device inventory and the current configurations (Part 1 does this for you).

Key devices

Device	Hostname	Role
Office PCs	PC1, PC2	Office users, behind the firewall (inside)
Access Switch	Access-Switch	Connects the Office PCs to the ASA (inside)
Edge Router	Edge Router	Connects the office to the ISP; routes to the Internet
ISP Router	ISP Router	Simulated ISP, needs a route back to the new 209.165.200.224/29 subnet
Public Server	Public Server	Simulated Internet: the test target for the PCs
ASA Firewall	ASA-FW	Firewall: inside / outside zones, PAT, ICMP inspection

Note: What changes: before the firewall the Edge Router does the PAT for the office. After the firewall the ASA does the PAT, and the Edge Router only routes.

2. Network plan (cheat sheet)

Network / link	Subnet	Addresses
Office LAN (inside)	192.168.1.0/24	ASA inside (VLAN 1) = 192.168.1.1, the gateway of the PCs (before the firewall this was the Edge Router G0/1)
ASA - Edge Router	209.165.200.224/29	Edge Router G0/1 = 209.165.200.225; ASA outside (VLAN 2) = 209.165.200.226
Edge Router - ISP	203.0.113.0/30	ISP Router = 203.0.113.1; Edge Router G0/0 = 203.0.113.2

Warning: The notes give the Public Server the address 209.165.200.226, which is also the ASA outside address. Two devices cannot use the same address, so check the Public Server address in the video and use that one. This guide writes "the Public Server address" in the tests.

ASA interfaces

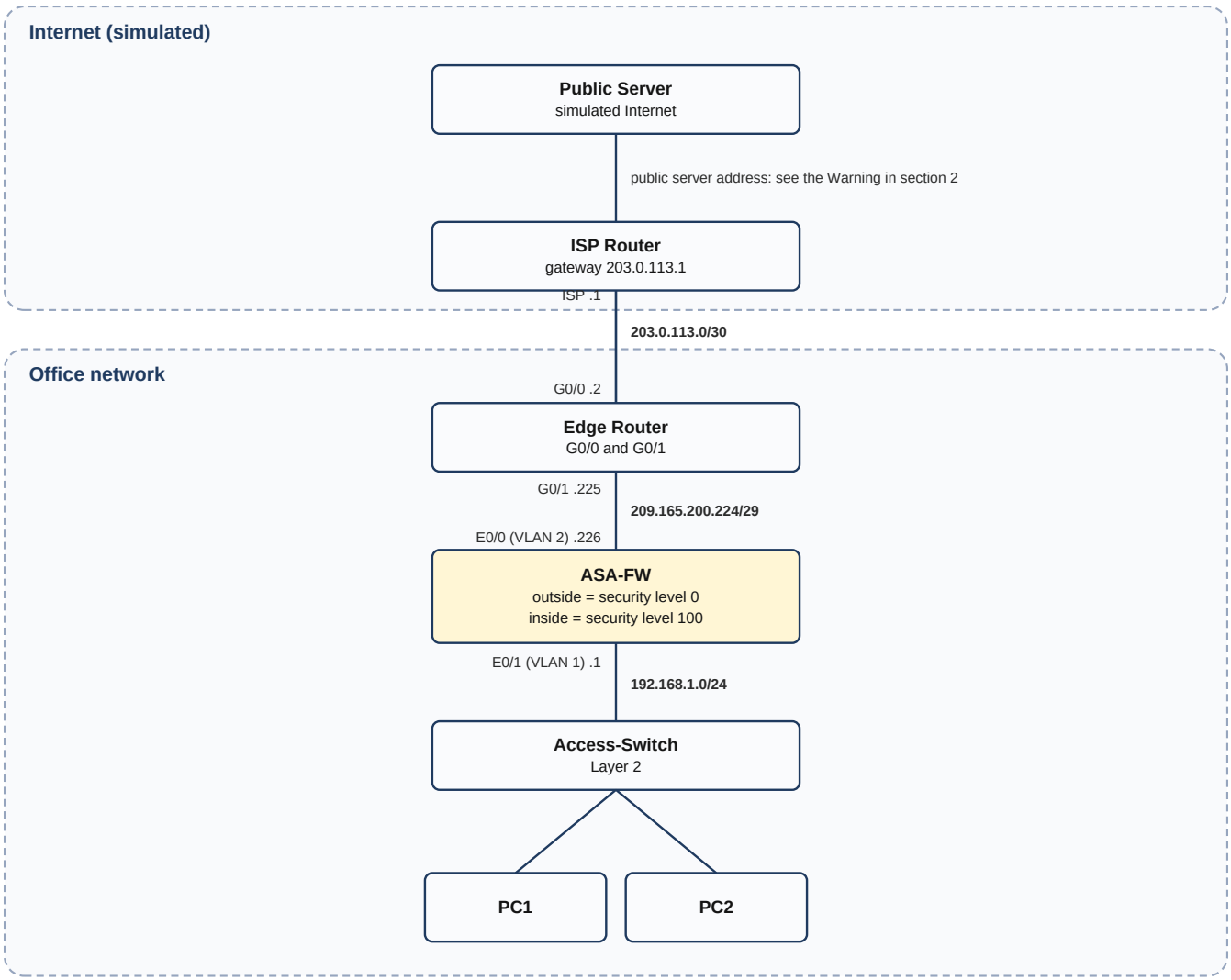
Port	VLAN	Name	Security level	IP address	Faces
E0/0	2	outside	0	209.165.200.226 /29	Edge Router G0/1
E0/1	1	inside	100	192.168.1.1 /24	Access Switch

Note: Traffic may go from a higher security level to a lower one (inside to outside). Unsolicited traffic from a lower level to a higher one (outside to inside) is blocked.

Before and after the firewall

Item	Before the ASA	After the ASA
Edge Router G0/1	192.168.1.1 /24 (inside NAT)	209.165.200.225 /29 (description: To ASA Firewall)
PAT / NAT	Edge Router (ACL 1, overload on G0/0)	ASA (network object, dynamic interface)
Gateway of the PCs	192.168.1.1 (Edge Router)	192.168.1.1 (ASA)
ISP Router routes	-	Static route to 209.165.200.224/29 via 203.0.113.2

Topology



The ASA (yellow) is new. It sits between the Edge Router and the Access Switch.

3. Physical connections

#	From	To	Cable
1	Edge Router G0/0	ISP Router	Ethernet
2	ISP Router	Public Server	Ethernet
3	Edge Router G0/1	ASA E0/0 (outside)	Ethernet
4	ASA E0/1 (inside)	Access-Switch	Ethernet
5	PC1	Access-Switch	Ethernet
6	PC2	Access-Switch	Ethernet

Note: Before the firewall, Edge Router G0/1 connected straight to the Access Switch. In Part 2 (step 4) you move that cable: Edge Router G0/1 goes to the ASA E0/0, and the ASA E0/1 goes to the Access Switch.

4. Part 1 - Review and test the current network

Step 1: See the current configuration on the Edge Router

Get familiar with the network before you change it. On the Edge Router:

```
enable
show ip interface brief
```

These interfaces are pre-configured:

Interface	IP address	Status
G0/0	203.0.113.2 /30	UP
G0/1	192.168.1.1 /24	UP

Routing table

```
show ip route
```

Type	Route	Meaning
Local	192.168.1.1/32 -> G0/1	LAN interface
Local	203.0.113.2/32 -> G0/0	WAN interface
Connected	192.168.1.0/24 -> G0/1	Internal LAN
Connected	203.0.113.0/30 -> G0/0	ISP connection
Static	0.0.0.0/0 -> 203.0.113.1 via G0/0	Default route to the ISP

Running configuration

```
show running-config
```

These things are already configured:

Feature	Configuration	What it does
NAT	G0/0 = ip nat outside; G0/1 = ip nat inside	Marks the outside and inside NAT boundary
PAT	ip nat inside source list 1 interface g0/0 overload	Translates many private IPs to the G0/0 public IP with different port numbers
ACL	access-list 1 permit 192.168.1.0 0.0.0.255	Defines which addresses are allowed translation

> **Note:** Your notes show the ACL network as 192.168.0.0, but the office LAN is 192.168.1.0 (see the appendix). Read the line in your own show running-config.

Step 2: Test connectivity

Everything must work before the firewall is added, so you know what "working" looks like.

Test	From	To	Expected
1	PC1	PC2	Ping replies
2	PC2	PC1	Ping replies
3	PC1	Public Server	Ping replies
4	PC2	Public Server	Ping replies

Step 3: Save the current configuration

Do this on the Edge Router and on the Access Switch.

```
end
copy running-config startup-config
```

5. Part 2 - Add the ASA and change the routers

Step 4: Modify the network structure

Add the ASA Firewall between the Edge Router and the Access Switch.

Action	Details
Remove	The cable between Edge Router G0/1 and the Access Switch
Add	ASA Firewall (hostname ASA-FW is set in Part 3)
Connect	Edge Router G0/1 to ASA E0/0 (outside)
Connect	ASA E0/1 (inside) to the Access Switch

Step 5: Remove the PAT rule and the ACL on the Edge Router

The ASA does the translation now, so the Edge Router must not translate any more. Remove the PAT rule first, then the ACL, then the NAT roles of the two interfaces.

```
enable
show running-config
configure terminal
no ip nat inside source list 1 interface g0/0 overload
no access-list 1
interface g0/0
 no ip nat outside
 exit
interface g0/1
 no ip nat inside
 exit
```

Step 6: New address for Edge Router G0/1

G0/1 now faces the ASA, in the new 209.165.200.224/29 subnet. Add a description too.

```
interface g0/1
 description ## To ASA Firewall ##
 ip address 209.165.200.225 255.255.255.248
 no shutdown
 exit
end
```

Note: G0/1 comes up when the ASA is connected and its E0/0 port is enabled (Part 3).

Step 7: Static route on the ISP Router

A new subnet now exists behind the Edge Router. The ISP Router needs a static route so it knows how to send the replies back to it.

```
enable
configure terminal
ip route 209.165.200.224 255.255.255.248 203.0.113.2
end
```

Save the configuration

Do this on the Edge Router and on the ISP Router.

```
copy running-config startup-config
```

6. Part 3 - ASA firewall configuration

Note: The Ethernet ports of the ASA act like switch ports: a port is assigned to a VLAN, and then the VLAN gets the IP address. Stay in configuration mode for steps 1 - 3.

Step 1: Ethernet ports and VLANs

Put the port facing the outside and the port facing the inside in separate VLANs and enable them.

```
enable
configure terminal
hostname ASA-FW
interface e0/0
 switchport access vlan 2
 no shutdown
 exit
interface e0/1
 switchport access vlan 1
 no shutdown
 exit
```

Port	VLAN	Faces
E0/0	2	outside (Edge Router)
E0/1	1	inside (Access Switch)

Step 2: Names, security levels, IP addresses and DHCP

Give each VLAN a logical name, a security level and an IP address. Then disable the DHCP daemon on the inside, so the ASA does not hand out addresses.

```
interface vlan 1
 nameif inside
 security-level 100
 ip address 192.168.1.1 255.255.255.0
 exit
interface vlan 2
 nameif outside
 security-level 0
 ip address 209.165.200.226 255.255.255.248
 exit
no dhcpd enable inside
```

Step 3: Default route on the firewall

The ASA needs to know where to send traffic that is not in its own networks: to the Edge Router.

```
route outside 0.0.0.0 0.0.0.0 209.165.200.225 1
```

Step 4: Test the link to the Edge Router

Ping the Edge Router from the ASA. Replies mean the outside link works.

```
ping 209.165.200.225
```

Step 5: PAT for the inside network

One network object translates the whole inside network to the ASA outside address.

```
object network Inside-Network
 subnet 192.168.1.0 255.255.255.0
 nat (inside,outside) dynamic interface
 exit
```

Line	What it does
subnet 192.168.1.0 255.255.255.0	Defines the inside network that is translated
nat (inside,outside) dynamic interface	PAT: many inside addresses share the outside interface address, with different ports

Step 6: Enable ICMP inspection

Without ICMP inspection the ASA treats an Echo Reply like normal traffic: the reply comes from the outside (level 0) toward the inside (level 100), and the main security rule is that unsolicited traffic from a less trusted network is not allowed. The ASA must understand that the ICMP reply belongs to an Echo request that was sent from the inside. ICMP inspection makes it keep track of the request and recognise the matching reply.

```
class-map inspection_default
 match default-inspection-traffic
 exit
policy-map global_policy
 class inspection_default
  inspect icmp
service-policy global_policy global
exit
```

Step 7: Confirm that PAT works

First ping the Public Server from each Office PC, then run:

```
enable
show xlate
show nat
```

Command	Look for
show xlate	Translations of the PC addresses (192.168.1.x) to the ASA outside address
show nat	The Inside-Network rule with translate hits

Step 8: Save the configurations

Save the ASA and the Edge Router.

```
ASA-FW:       write memory
Edge Router:  copy running-config startup-config
```

7. End devices

Device	IP address	Gateway
PC1, PC2	Same 192.168.1.x settings as before the firewall	192.168.1.1 (now the ASA)
Public Server	The address from the video (see the Warning in section 2)	ISP side

Note: The ASA inside address is the same 192.168.1.1 that the Edge Router G0/1 had, so the PCs keep their default gateway and need no change.

8. Quick verification

Check	Command / test
Edge Router interfaces	show ip interface brief (G0/0 = 203.0.113.2, G0/1 = 209.165.200.225, both up)
Edge Router is clean	show running-config (no ip nat lines and no access-list 1)
Route back on the ISP	show ip route on the ISP Router (static route 209.165.200.224/29 via 203.0.113.2)
ASA interfaces	show interface ip brief (VLAN 1 = 192.168.1.1, VLAN 2 = 209.165.200.226)
ASA names and levels	show nameif (inside = 100, outside = 0)
ASA default route	show route (default via 209.165.200.225)
ASA to Edge Router	ping 209.165.200.225
ICMP inspection	show running-config policy-map (inspect icmp is listed)
PAT rule and translations	show nat | show xlate
PC to PC	Ping PC2 from PC1 and PC1 from PC2
PC to Internet	Ping the Public Server from each Office PC (replies)

9. Troubleshooting: PCs cannot ping the Public Server

The ASA can ping the Edge Router, but the Office PCs get no replies from the Public Server.

Where	Command	Look for
ASA-FW	ping 209.165.200.225	Replies: the outside link is fine
ASA-FW	show nat	The Inside-Network PAT rule exists
ASA-FW	show route	Default route via 209.165.200.225
ASA-FW	show running-config policy-map	inspect icmp is listed under inspection_default
ISP Router	show ip route	Static route to 209.165.200.224/29 via 203.0.113.2

Problem found	Why it matters
ICMP inspection is not enabled on the ASA	The Echo Reply comes from the outside (level 0) toward the inside (level 100) and is treated as unsolicited traffic from a less trusted network. With inspect icmp the ASA remembers the Echo request and accepts the matching reply.
No static route on the ISP Router	The ISP Router does not know the 209.165.200.224/29 subnet behind the Edge Router, so the replies cannot find their way back.

Fix

Enable ICMP inspection (Part 3, step 6), or add the ISP route (Part 2, step 7), then ping the Public Server again from each PC.

Appendix: corrections made to the original notes

In the notes	Used in this guide
'switchport mode access vlan 2' / 'vlan 1' (ASA ports)	switchport access vlan 2 / switchport access vlan 1
'ip route 20 9.165.200.224 ...'	ip route 209.165.200.224 255.255.255.248 203.0.113.2
'no shutdown exit' on one line	no shutdown and exit on separate lines
'match default-inspection traffic'	match default-inspection-traffic
ASA steps numbered 1-5, then 7, 8, 9	Steps renumbered 1-8
'configure terminal' before 'show xlate' / 'show nat'	Not needed: run them from the enable prompt
'show run' inside configuration mode (Edge Router, step 5)	show running-config first, then configure terminal
'access-list 1 permit 192.168.0.0 0.0.0.255'	The office LAN is 192.168.1.0: read your own show running-config
Public Server 209.165.200.226 (same as the ASA outside address)	Use the Public Server address from the video
Typos: 'thr inside', 'Confgiure', 'procced', 'rue'	Corrected

Added so the commands work (not written in the notes): 'write memory' for the ASA, the before-and-after table, the verification commands and the troubleshooting section.